A

Mini Project Report

On

BASED BOILERS IOT TEMPERATURE AND PRESSURE MONITORING SYSTEM

THIS PROJECT REPORT IS SUBMITTED
TO

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

IN PARTIAL FULLFILMENT OF THEN REQUIREMENT FOR THE AWARD OF

B-TECH
In

ELECTRONICS&COMMUNICATIONENGINEERING

BY

G.RAMYA	(23681A0412)
G.VANAJA	(24685A0404)
K.SAI PRANAY	(24685A0405)
T.HARSHITH KUMAR	(23681A0428)

Under the Esteemed Guidance of

Mr.B.HANUMANTHU

CHRISTU JYOTHI INSTITUTE OF TECHNOLOGY AND SCIENCE

NBA &NAAC Accredited, Permanently Affiliated to JNTUH
Colombo, nagar Yeshwanthapur, Jangaon, 50616



DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING
2025-2026

CHRISTU JYOTHI INSTITUTE OF TECHNOLOGY AND SCIENCE

NBA &NAAC Accredited, Permanently Affiliated to JNTUH

Colombo Nagar, Yeshwanthapur, Jangaon,506167

B-TECH IN ELECTRONICS & COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the report entitled ON**" IOT-BASED BOILERS TEMPERATURE AND PRESSURE MONITORING SYSTEM "**submitted by, G.RAMYA (23681A0412),G.VANAJA (24685A0404),K.SAI PRANAY (24685A0405),T.HARSHITH KUMAR(23681A0428).Impartial fulfillment for the requirements of the degree of Bachelor Of Technoogy **"ELECTRONICS & COMMUNICATION ENGINEERING"** and submitted to Department of Electronic & Communication Engineering and submitted to the Department Of Electronics &Communication Engineering of **Christu Jyothi Institute of Technology and Science**, Jangaon. This is to certify that the above statement made by the candidate is correct to the best of my knowledge

INTERNAL GUIDE

B. HANUMANTHU

ASSISTANT PROFESSOR

H.O.D

ALLANKI SANYASI RAO

INSTITUTE VISION AND MISSION

VISION

To admit and groom students from rural background sand be a truly rural Technical institute benefiting, society and nation as a whole institute.

MISSION

1. The mission of the institution is to create, deliver and refine knowledge. Being a rural technical institute, our mission is to.
2. Enhance our position to one of the best technical institutions and measure our performance against the highest defined standards.
3. Provide high quality learning environment for our students for their greater well-being to equip them with highest technical and professional ethics.

DEPARTMENT VISION MISSION

VISION

To be an established center of excellence in Electronics and Communication Engineering facilitating youth towards professional, leadership and industrial needs.

MISSION

1. To impart quality education in electronics and communication engineering through innovative teaching and hands-on training.
2. To encourage students to engage in lifelong learning, research, and development to meet industry and societal needs.
3. To promote ethical and professional values while preparing students for successful careers and responsible citizenship.
4. To establish partnerships with industry and academics for mutual growth and advancement in technology.

DECLARATION

We here by declare that the project report entitled "IOT-BASED BOILERS TEMPERATURE AND PRESSURE MONITORING SYSTEM"Which us being submitted as Mini Project in Electronics and communication to Christu Jyothi Institute of Technology & Science, is a record of an original work done by me under the guidance of Mr. B.Hanumanthu, Assistant professor of ECE Dept , CJITS.

G.RAMYA	(23681A0412)
G.VANAJA	(24685A0404)
K.SAI PRANAY	(24685A0405)
T.HARSHITH KUMAR	(23681A0428)

ACKNOWLEDGEMENT

We would take immense pleasure to acknowledge with gratitude; the help and support extended during the course of our project entitled **"IOT-BASED BOILERS TEMPERATURE AND PRESSURE MONITORING SYSTEM"** from people who have helped in the successful completion of this project all.We are heartily thankful to the Director **Rev. Fr. VIJAYPAUL REDDY,** for giving this opportunity.

We express thanks to our Principal **Dr.S.CHANDRASHEKAR REDDY,** for giving this opportunity.We express thanks and gratitude to **A.SANYASI RAO, DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING** for his encouraging support and guidance throughout the project.We take this opportunity to express our sincere thanksto our **project guide, Mr.**B.Hanumanthu for his exemplary guidance and monitoring throughout the course of this project. Without the help and guidance given by him time to time, we couldn't have completed the project. We would like to express our special gratitude and thanks to all who extended their helping hand in accomplishment of the project.

Thanking you,

G.RAMYA (23681A0412)
G.VANAJA (24685A0404)
K.SAI PRANAY (24685A0405)
T.HARSHITH KUMAR (23681A0428)

ABSTRACT

The IoT-based Boiler Temperature and Pressure Monitoring System is an advanced real-time monitoring solution designed to improve the safety, efficiency, and reliability of industrial boiler operations. Boilers are widely used in industries such as power plants, manufacturing units, food processing industries, and chemical plants for steam generation and heating applications. Continuous monitoring of boiler parameters such as temperature and pressure is critical because abnormal variations may lead to equipment failure, reduced efficiency, or even hazardous accidents like boiler explosions.

Traditional monitoring methods often rely on manual inspection or standalone control systems, which may not provide real-time remote access or early warning alerts. To overcome these limitations, the proposed system integrates Internet of Things (IoT) technology to enable continuous and remote monitoring.

In this system, temperature sensors (such as DS18B20 or thermocouples) and pressure sensors are used to measure the boiler's internal conditions. These sensors are connected to a microcontroller such as ESP8266, ESP32, or Arduino. The microcontroller collects real-time data application or mobile application, allowing operators to monitor boiler performance from anywhere in the world. If the temperature or pressure exceeds predefined safe limits, the system automatically generates alerts in the form of notifications, SMS, or emails to ensure immediate action can be taken.

The system not only enhances safety by providing early fault detection but also improves operational efficiency by enabling predictive maintenance. Historical data stored in the cloud can be analyzed to detect trends and prevent future failures. Additionally, automation features such as automatic shutdown or pressure release control can be integrated to further enhance protection..

TABLE OF CONTENTS

CHAPTER NUMBER	DESCRIPTION	PAGE NUMBRES
1.0	INTRODUCTION	1-3
1.1	HOW IT WORKS	4
1.2	OVER VIEW OF IOT-BASED BOILERS TEMPERATURE AND PRESSURE MONITORING SYSTEM	5
1.3	OBJECTIVE	6
2	LITERATURE	7-8
3.0	INTRODUCTION TO IOT	9
3.1	CHARACTERITICS OF IOT	10-11
3.2	APPLICATIONS OF IOT	11-13
4	HARDWARE COMPONENTS	14-26
5	CIRCUIT DIAGRAM	27
6	WORKING PRINCIPLE	29
7	OPERATION	28
8.0	IMPLEMENTATION AND TESTING	29
8.1	MERITS	30-31
8.2	DEMERITS	32
9.0	RESULT	34
9.1	APPLICATIONS	35
9.2	PREVENTION TECHNIQUES	36
10.0	CONCLUSION	37
10.1	REFERENCE	38
10.2	APPENDICES	39-41

LIST OF FIGURES

Figure1.1–Microcontroller ESP32 Module of iot based boiler temperature and pressure monitoring system

Figure1.2–Pin diagram

Figure1.3–Temperature Sensor (DHT22)

Figure1.4–Pressure Sensor (Potentiometer)

Figure 1.5–Pin Display OLED

Figure1.6–Buzzer

Figure1.7–Connecting Wires

Figure1.8–Circuit diagram

CHAPTER-1

1.0 INTRODUCTION

An IoT-based boiler temperature and pressure monitoring system using the ESP-32 module is developed to improve the safety, reliability, and efficiency of boiler operations. Boilers are commonly used in industrial and domestic applications, and maintaining proper temperature and pressure levels is very important to avoid accidents and equipment damage. Traditional boiler monitoring methods require continuous manual supervision, which is time-consuming and may lead to human errors.

In this system, a temperature sensor is used to continuously measure the boiler temperature, while a potentiometer is employed to represent or control pressure variations. These sensor values are collected and processed by the ESP-32(ESP8266) Wi-Fi module, which acts as the core controller and enables internet connectivity. A 4-pin display is used to show real-time temperature and pressure values locally, allowing easy on-site monitoring. For safety purposes, a buzzer is included to provide an audible alert whenever the measured values exceed predefined safe limits.

The ESP-32 sends the collected data to an IoT platform, enabling remote monitoring through a mobile or web application. This allows users to observe boiler conditions from anywhere and take timely action in case of abnormal conditions. Overall, the system offers a low-cost, efficient, and smart solution for real-time boiler monitoring, enhanced safety, and reduced manual intervention.An IoT-based boiler temperature and pressure monitoring system is a modern industrial solution designed to enhance safety, efficiency, and reliability by replacing manual, reactive monitoring with real-time, automated data acquisition.

By integrating sensors (such as LM35 for temperature and piezo-resistive transducers for pressure) with smart microcontrollers like ESP32 or NodeMCU, the system continuously gathers critical operational data and transmits it securely via Wi-Fi or cellular networks to a cloud platform. Operators can access this live data on a mobile app or web dashboard, allowing them to monitor conditions from remote locations, receive instant alerts when parameters exceed safe thresholds, and trigger automated safety shutdowns, which significantly reduces the risk of explosions and hazardous accidents.

Furthermore, the system logs historical data to enable predictive maintenance, ensuring optimal

combustion, reducing energy consumption, and extending the overall lifespan of the equipment.

An IoT-based boiler temperature and pressure monitoring system uses sensors and a microcontroller (like ESP32) to measure boiler conditions in real time. The data is sent over the internet for remote monitoring. It helps detect abnormal temperature or pressure and provides alerts, improving safety and efficiency.

An IoT-based boiler temperature and pressure monitoring system is a smart solution designed to continuously track and control critical boiler parameters using internet-connected devices. In industrial boilers, maintaining proper temperature and pressure is essential for safe and efficient operation. Any abnormal rise or fall can lead to equipment damage, energy loss, or even hazardous accidents.

In this system, sensors such as temperature (e.g., DHT22) and pressure sensors are connected to a microcontroller like ESP32. These sensors collect real-time data from the boiler and send it to the cloud through the internet. The data can be monitored remotely using a mobile app or web dashboard. If the readings exceed safe limits, the system can trigger alerts through a buzzer or notifications, enabling quick action.

An IoT-based boiler temperature and pressure monitoring system is designed to automatically measure and supervise important boiler parameters using smart electronic devices. Boilers are widely used in industries, and maintaining proper temperature and pressure is essential for safe and efficient operation. Manual monitoring can be difficult and may lead to errors, so an automated system helps in continuous observation.In this system, sensors such as temperature sensors (like DHT22) and pressure sensors are connected to a microcontroller such as ESP32. These sensors collect real-time data from the boiler environment. The microcontroller processes this data and sends it to cloud platforms through the internet, enabling users to access the information from anywhere.

The system also includes display units (like LCD or OLED) and alert mechanisms such as buzzers. If the temperature or pressure exceeds safe limits, the system immediately notifies the user through alarms or mobile alerts. This helps in taking quick corrective actions and prevents accidents. An IoT-based boiler temperature and pressure monitoring system is an advanced solution that integrates sensors, microcontrollers, and internet connectivity to ensure safe boiler operation. Boilers are critical components in many industries such as power plants, manufacturing, and chemical processing, where maintaining optimal temperature and pressure

is necessary to avoid failures and accidents.

The system uses sensors to continuously measure temperature and pressure inside the boiler. These sensors send analog or digital signals to a microcontroller like ESP32, which processes the data. The microcontroller is programmed to compare the measured values with predefined safe limits to determine whether the system is operating normally.

One of the key features of this system is real-time data transmission. Using Wi-Fi or other communication technologies, the ESP32 sends the collected data to cloud platforms or IoT dashboards. This allows operators to monitor boiler conditions remotely using smartphones, laptops, or computers without being physically present near the boiler.

In addition to monitoring, the system provides instant alerts when abnormal conditions occur. For example, if the temperature or pressure exceeds the safe threshold, a buzzer is activated, and notifications can be sent to users via mobile applications or emails. This early warning system helps in preventing equipment damage and ensures the safety of workers.

Furthermore, the system can store historical data, which is useful for analysis and decision-making. By studying trends in temperature and pressure, industries can perform predictive maintenance, reduce downtime, and improve overall efficiency. This also helps in optimizing fuel consumption and reducing operational costs.

1.1 How its works:

1.Power Supply Initialization: The system is powered using a regulated power supply (such as 5V or 3.3V). This power is provided to the microcontroller (ESP32 or ESP8266) and all connected sensors.

2.Sensor Data Collection: A temperature sensor (such as LM35, DS18B20, or DHT11) is used to measure the boiler temperature. A pressure sensor is used to monitor the pressure inside the boiler. These sensors continuously sense real-time temperature and pressure values.

3.Microcontroller Processing: The sensors are connected to the ESP32/ESP8266 microcontroller. The microcontroller reads the analog or digital data from the sensors.
The raw sensor values are converted into meaningful temperature (°C) and pressure (bar or psi) readings.

4.Wi-Fi Connectivity: The ESP32/ESP8266 connects to a Wi-Fi network.
Once connected, it sends the sensor data to the Blynk cloud server using the internet.

5.Data Display on Blynk App : The Blynk mobile application displays the real-time temperature and pressure values using widgets such as gauges, LCD displays, or graphs.
Users can monitor the boiler condition from anywhere using a smartphone.

6.Threshold Monitoring and Alerts : Safe limit values for temperature and pressure are predefined in the program. If the temperature or pressure exceeds the safe limit,
Blynk sends a notification alert to the user's mobile phone.

1.2 Overview of IOT Based Boilers Temperature and Pressure Monitoring System

To design and develop an IoT-based system for real-time monitoring of boiler temperature and pressure.

To use the ESP-32 module for data processing and internet connectivity.

To measure boiler temperature using a suitable temperature sensor accurately.

To simulate and monitor pressure levels using a potentiometer.

To display real-time temperature and pressure values on a 4-pin OLED/LCD display.

To provide an audio alert using a buzzer when temperature or pressure exceeds safe limits.

To enable remote monitoring of boiler parameters through an IoT platform.

To improve boiler safety, reduce manual monitoring, and prevent accident

1.3 OBJECTIVES

Real-Time Monitoring of Boiler Parameters: The primary objective is to continuously monitor the temperature and pressure of the boiler. The temperature sensor collects accurate readings, while the potentiometer can simulate or adjust pressure readings. This ensures that the boiler operates within safe limits at all times.

Remote Access Using IoT: By using the ESP-12E module, the system connects to the internet. Data is transmitted to an IoT platform such as Blynk, enabling users to monitor the boiler status remotely via smartphones or web applications.

Immediate Alerts for Safety: The buzzer acts as an alert system. When the temperature or pressure exceeds predefined safe limits, the buzzer sounds immediately, notifying operators to take corrective action. This helps prevent accidents, damage, or boiler failure.

Local Display of Data: The 4-pin display (like an OLED or LCD) shows real-time readings of temperature and pressure on the spot. This allows users to quickly check the boiler status without relying solely on remote monitoring.

Control and Simulation Using Potentiometer: The potentiometer is used to simulate changes in pressure or as a manual adjustment tool. This helps in testing the system, calibrating sensors, and understanding how the boiler reacts to parameter variations.

Reduce Manual Monitoring: One major goal is to automate the monitoring process, reducing the need for manual checking of boiler parameters, saving time, and minimizing human errors.

CHAPTER-2

LITERATURE REVIEW

In recent years, Internet of Things (IoT) technologies have increasingly been adopted for real-time monitoring and control in industrial applications due to their ability to provide remote access, continuous data logging, and alerting capabilities. IoT platforms such as ThingSpeak allow sensor data to be transmitted over Wi-Fi from embedded devices and visualized on the cloud for remote monitoring and analysis. thingspeak.mathworks.com

IoT in Boiler and Industrial Monitoring

Several studies highlight the use of IoT to monitor thermal and pressure parameters in industrial environments. For example, researchers developed an IoT-based real-time and remote boiler fuel monitoring system in a production plant. In this work, sensors for temperature, pressure, fuel level, and humidity were interconnected with a microcontroller and data was sent to the ThingSpeak platform for visualization and analysis. The system improved monitoring accuracy and reduced manual inefficiencies compared to traditional inspections. ResearchGate

Similarly, IoT systems have been designed to monitor boiler parameters such as temperature and water levels, showing that real-time sensor data transmission to cloud platforms enhances safety and operational insight in thermal systems. These studies emphasize that remote monitoring via IoT significantly improves decision-making and emergency response, particularly where manual measurement is slow or unsafe. IJRST Journal

ThingSpeak as a Cloud Platform

ThingSpeak is an open-source IoT analytics platform that provides sensor data collection, real-time visualization, and analysis tools, making it popular for academic and industrial prototype systems. It supports direct data uploads from devices like ESP8266/ESP32 via Wi-Fi and offers built-in graphing and data processing features compatible with MATLAB for deeper analysis. □

thingspeak.mathworks.com +1

In many IoT projects involving temperature and pressure logging, ThingSpeak is often used to aggregate data in cloud channels, which can then be accessed through dashboards or apps to observe trends over time—important for monitoring systems where parameter drift could indicate faults or risks.

thingspeak.mathworks.com

Sensor Integration in IoT Systems

Literature on IoT monitoring highlights the role of digital sensors (e.g., DHT22 for temperature and humidity) and pressure sensors that interface with microcontrollers like ESP8266. These sensors collect environmental and system data, which is then transmitted to cloud services for logging and visualization. Studies indicate that IoT implementations provide continuous automated monitoring compared to conventional data collection, enabling predictive alerts and better system control.

Several research works highlight the use of temperature sensors like DHT11/DHT22 and pressure sensors integrated with microcontrollers such as Arduino, ESP8266, and ESP32. These systems collect real-time data and transmit it to cloud platforms using Wi-Fi modules, enabling remote monitoring through web dashboards or mobile applications. Researchers have emphasized the importance of continuous monitoring to prevent hazardous situations like boiler explosions caused by excessive temperature or pressure.

Many studies also focus on the use of IoT platforms such as ThingSpeak, Blynk, and Firebase, which allow data visualization, storage, and analysis. These platforms help in tracking historical data trends and predicting potential failures. Some advanced research incorporates machine learning techniques for predictive maintenance, where the system can forecast abnormal conditions before they occur.

In addition, several papers discuss the implementation of alert systems, such as buzzers, SMS notifications, or email alerts, to inform operators when parameters exceed safe limits. This reduces response time and enhances safety. Researchers have also worked on improving system reliability, reducing power consumption, and making the design cost-effective for small-scale industries

CHAPTER-3

INTERNET OF THINGS

3.0 INTRODUCTION TO INTERNET OF THINGS

The Internet of Things (IoT) refers to a network of physical devices embedded with sensors, software, and communication technologies that enable them to collect and exchange data over the internet. These devices can range from simple household appliances to complex industrial machines. IoT allows real-time monitoring, automation, and control of systems without direct human intervention. By connecting devices and enabling data sharing, IoT improves efficiency, accuracy, and decision-making in various fields such as healthcare, agriculture, smart homes, and industrial automation.

The Internet of Things (IoT) refers to a network of physical devices embedded with sensors, software, and communication technologies that enable them to collect and exchange data over the internet. These devices can range from simple household appliances to complex industrial machines. IoT allows real-time monitoring, automation, and control of systems without direct human intervention. By connecting devices and enabling data sharing, IoT improves efficiency, accuracy, and decision-making in various fields such as healthcare, agriculture, smart homes, and industrial automation. The Internet of Things (IoT) is a rapidly growing technology that enables everyday objects to connect to the internet and communicate with each other. These objects are equipped with sensors, processors, and communication modules that allow them to collect, send, and receive data. This connectivity helps devices operate intelligently by responding to real-time conditions and user requirements.

IoT works through a combination of hardware and software components. Sensors gather data such as temperature, humidity, motion, or pressure, while actuators perform actions based on the processed data. The data is transmitted through communication networks like Wi-Fi, Bluetooth, or cellular networks to cloud platforms or local servers, where it is analyzed and stored. Users can then access this information through applications or dashboards for monitoring and control.

3.1 CHARACTERISTICS OF IoT

Connectivity: Connectivity is the backbone of IoT. Devices such as sensors, microcontrollers, and actuators are connected through the internet or local networks. This allows seamless communication between devices and enables remote monitoring and control from anywhere in the world.

Sensing and Data Collection: IoT devices are equipped with sensors that continuously collect data from the environment. These sensors can measure parameters such as temperature, pressure, humidity, light, and motion. The collected data forms the basis for analysis and decision-making.

Intelligence and Data Processing: IoT systems are capable of analyzing the collected data using embedded processors or cloud computing. This intelligence allows the system to make decisions, identify patterns, and perform predictive analysis, often without human intervention.

Automation and Control: Automation is a key feature of IoT. Based on the analyzed data, IoT systems can automatically perform actions such as turning devices on or off, triggering alarms, or adjusting system parameters. This reduces human effort and increases efficiency.

Real-Time Operation: IoT systems operate in real time, meaning data is collected, transmitted, and processed instantly. This is crucial for applications like industrial monitoring, healthcare systems, and smart homes where immediate responses are required.

Scalability: IoT systems are highly scalable. New devices can be added to the network without affecting the existing system. This flexibility makes IoT suitable for both small-scale and large-scale applications.

Heterogeneity: IoT networks consist of a wide variety of devices with different hardware, software, and communication protocols. Despite these differences, they are designed to work together efficiently.

Security and Privacy: Since IoT devices handle sensitive data and are connected to the internet, security is a major concern. Proper measures such as encryption, authentication, and secure communication protocols are necessary to protect data and prevent unauthorized access.

Energy Efficiency: Many IoT devices are designed to consume low power, especially those that operate on batteries. Efficient energy usage ensures long-term operation without frequent maintenance.

3.2 APPLICATIONS OF IoT

Smart Homes: IoT is widely used in smart home systems to automate and control household appliances. Devices such as smart lights, thermostats, fans, security cameras, and door locks can be controlled remotely using smartphones or voice assistants. IoT helps in energy saving, improved security, and increased convenience. For example, a smart thermostat can automatically adjust room temperature based on user preferences.

Healthcare (Smart Healthcare): In healthcare, IoT plays a crucial role in patient monitoring and management. Wearable devices like fitness bands and smartwatches track heart rate, blood pressure, and oxygen levels. Hospitals use IoT-based systems for remote patient monitoring, smart beds, and real-time health tracking. This helps doctors provide timely treatment and reduces hospital visits.

Industrial IoT (IoT): Industries use IoT to improve efficiency and productivity. Sensors are used in machines to monitor temperature, pressure, vibration, and performance. This helps in predictive maintenance, reducing machine breakdowns, and improving safety. For example, in a boiler monitoring system, IoT sensors track temperature and pressure continuously.

Smart Cities: IoT is used in developing smart cities by improving urban infrastructure. Applications include smart street lighting, traffic management, waste management, and water supply systems. Sensors collect real-time data to reduce congestion, save energy, and improve public services. For instance, smart traffic signals adjust timing based on vehicle density.

Agriculture (Smart Farming): IoT helps farmers increase crop yield and reduce wastage. Sensors monitor soil moisture, temperature, humidity, and weather conditions. Automated irrigation systems supply water only when needed. Farmers can monitor their fields remotely and make better decisions regarding fertilization and harvesting.

Transportation and Logistics: IoT improves transportation systems by enabling vehicle tracking, route optimization, and fleet management. GPS-enabled IoT devices help track goods in real-time. It ensures timely delivery and reduces fuel consumption. Smart traffic systems also help in reducing accidents and congestion.

Retail Industry: In retail, IoT is used for inventory management, smart shelves, and customer behavior analysis. Sensors track product availability and automatically update stock levels. Retailers can provide personalized shopping experiences based on customer preferences and buying patterns.

Energy Management: IoT helps in efficient energy usage by monitoring and controlling electricity consumption. Smart meters track energy usage in real-time and help reduce wastage. Renewable energy systems like solar panels also use IoT for performance monitoring and optimization.

Environmental Monitoring: IoT is used to monitor environmental conditions such as air quality, water quality, temperature, and pollution levels. Sensors collect real-time data, which helps in taking preventive measures to protect the environment. For example, IoT systems can detect harmful gas leaks or rising pollution levels.

Security and Surveillance: IoT enhances security systems through smart cameras, motion sensors, and alarm systems. These devices can detect unusual activities and send alerts to users instantly. It is widely used in homes, offices, banks, and public places for safety.

Wearable Technology: Wearable IoT devices like smartwatches, fitness trackers, and smart glasses help in tracking health, fitness, and daily activities. These devices provide real-time feedback and help users maintain a healthy lifestyle.

Smart Education: IoT improves education by enabling smart classrooms with connected devices like smart boards, projectors, and attendance systems. Teachers can monitor student performance, and students can access learning materials anytime, anywhere.

Smart Grid: IoT is used in smart electrical grids to monitor and manage electricity distribution efficiently. Smart grids use sensors and smart meters to detect faults, reduce power loss, and balance supply and demand. This ensures reliable and uninterrupted power supply.

CHAPTER-4

HARDWARE COMPONENTS

1.Microcontroller ESP32 Module

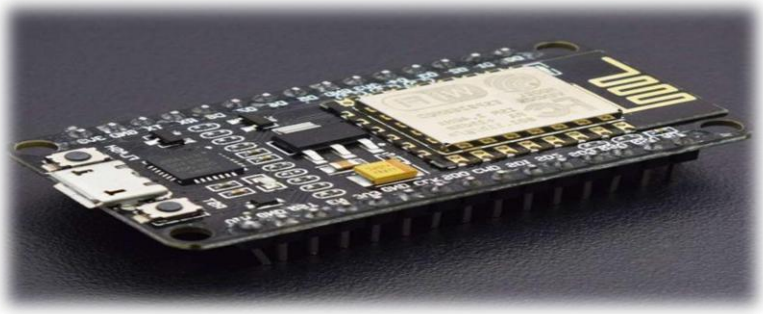


Figure1.1–Microcontroller ESP32 Module of iot based boiler temperature and pressure monitoring system

The ESP-32 is a Wi-Fi module based on the popular ESP8266EX microcontroller, widely used in Internet of Things (IoT) applications. It is a compact, low-cost module that integrates a 32-bit Tensilica L106 RISC processor along with a full TCP/IP protocol stack, allowing it to connect directly to Wi-Fi networks without the need for an external microcontroller. The module operates at 3.3V and typically comes with 4MB of flash memory, which is used to store programs and data. It supports the IEEE 802.11 b/g/n wireless standards and can function in different modes such as Station mode (connecting to a Wi-Fi router), Access Point mode (creating its own Wi-Fi network), or both simultaneously. The ESP-12E provides multiple GPIO pins for interfacing with sensors, relays, displays, and other peripherals, and it supports communication protocols such as UART, SPI, I2C (software-based), PWM, and ADC. One analog input pin is available with a voltage range of 0–1V, which is used for reading analog sensor data. Important pins include VCC, GND, TX, RX, EN (chip enable), RST (reset), and GPIO0 (used for flashing firmware). The module can be programmed using the Arduino IDE, NodeMCU firmware, or AT commands, making it very flexible for developers. Due to its built-in Wi-Fi capability, small size, low power consumption, and affordable price, the ESP-12E is extensively used in smart home automation systems, wireless monitoring systems, IoT-based temperature and pressure monitoring projects, and remote control applications. However, it is not 5V tolerant and must be powered carefully with a stable 3.3V supply.

USES

The ESP32 microcontroller is widely used in various applications due to its built-in Wi-Fi and Bluetooth capabilities, high processing speed, and low power consumption. It plays a major role in Internet of Things (IoT) systems by enabling devices to connect to the internet, collect data from sensors, and transmit it to cloud platforms for monitoring and control. In home automation, ESP32 is used to control appliances like lights, fans, and air conditioners remotely through mobile apps or voice commands. It is also used in industrial automation for monitoring parameters such as temperature, pressure, and humidity to improve efficiency and safety. Additionally, ESP32 is applied in wireless sensor networks, smart agriculture systems for irrigation and crop monitoring, and wearable devices like fitness trackers. It is commonly used in robotics, security systems, GPS tracking, and data logging applications. Due to its versatility and cost-effectiveness, ESP32 is also popular in educational projects and DIY electronics for learning embedded systems and IoT development.

PIN DIAGRAM

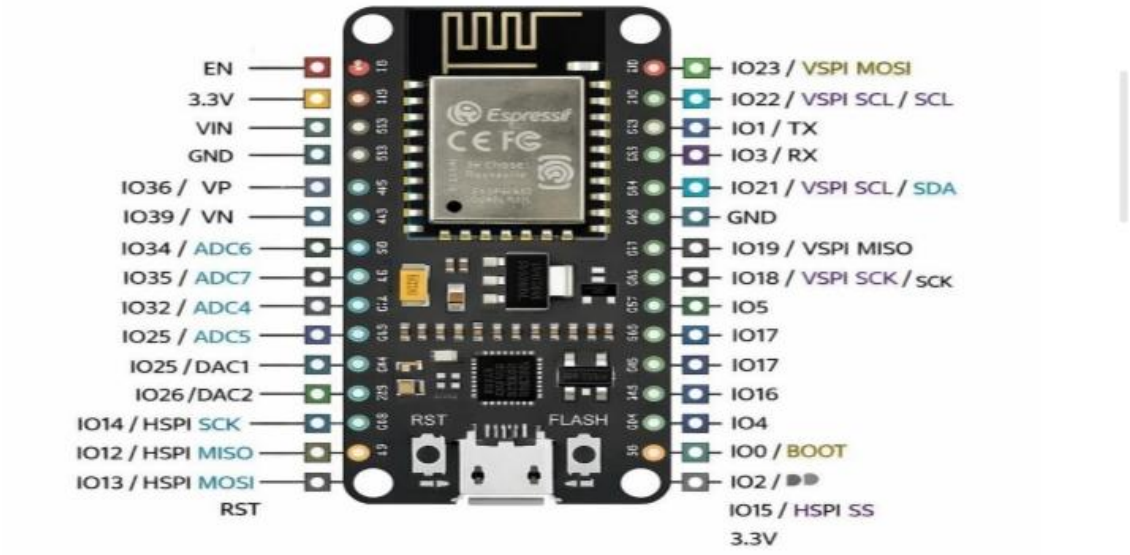


Figure1.2–Pin diagram

PIN DESCRIPTION

➢ Analog Pins : Read Analog signals from sensors.

➢ Digital Pins : Used to control LEDs, relays, and other devices.

➢ Communication Pins : Support protocols like UART, SPI, and I2C.

➢ PWM Pins : Used for motor control and LED brightness.

➢ Touch Pins : Support capacitive touch sensing.

➢ Power Pins: Provide power to the board (3.3V, GND).

GPIO Pins : Used for input and output connections.

Wi-Fi Modes

• Station Mode (STA) – Connects to a Wi-Fi network.

• Access Point Mode (AP) – Creates its own Wi-Fi network.

• STA + AP Mode – Connects and creates network at the same time.

• Promiscuous Mode – Monitors Wi-Fi network traffic.

ARCHITECTURE

➢ Processor: 32-bit Xtensa LX6 dual-core microprocessor

➢ Clock Speed: Up to 240 MHz

➢ Wi-Fi Stack: Integrated TCP/IP protocol stack

➢ Operating Voltage: 3.3 V

➢ Flash Memory: Up to 16 MB (varies by module)

➢ RAM: 520 KB SRAM

Features of ESP32

➢ Built-in Wi-Fi (802.11 b/g/n) and Bluetooth (Classic + BLE) support

➢ Dual-core processor for faster performance

➢ Supports TCP/IP protocol stack

➢ Multiple GPIO pins with PWM support

➢ Supports UART, SPI, and I2C communication

➢ Low power consumption modes

➢ Easy programming using Arduino IDE, Micro Python, and ESP-IDF

2.Temperature Sensor (DHT22)

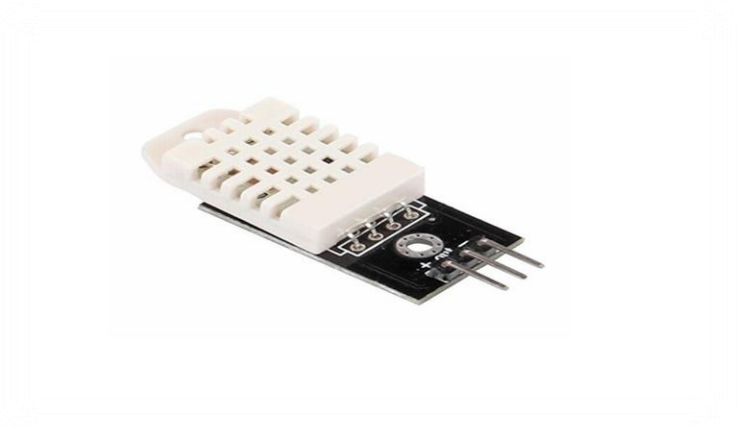


Figure1.3–Temperature Sensor (DHT22)

The DHT22 (also known as AM2302) is a widely used digital temperature and humidity sensor designed for accurate environmental monitoring in embedded systems and IoT applications. It measures temperature in the range of –40°C to 80°C with an accuracy of ±0.5°C, and humidity from 0% to 100% relative humidity (RH) with an accuracy of ±2–5% RH, making it more precise than the commonly used DHT11 sensor. The DHT22 operates on a supply voltage of 3.3V to 6V and provides a calibrated digital output signal, which means no external analog-to-digital converter (ADC) is required. Internally, it consists of a capacitive humidity sensing element, a thermistor for temperature measurement, and a built-in microcontroller that processes and converts the sensed analog signals into digital data.

The sensor communicates using a single-wire serial communication protocol, transmitting a total of 40 bits of data, which include humidity data (16 bits), temperature data (16 bits), and a checksum (8 bits) for error detection. One important characteristic of the DHT22 is its sampling rate, which is one reading every 2 seconds (0.5 Hz), making it suitable for applications that do not require very fast measurements. It has four pins: VCC (power supply), Data (serial output), NC (not connected), and GND (ground), and typically requires a 10kΩ pull-up resistor between VCC and the Data pin for proper communication.

PINS OUT OF TEMPERATURE SENSOR

3-Pin Temperature Sensor (e.g., LM35, DHT11 basic)

- VCC – Power supply (3.3V or 5V)
- GND – Ground
- OUT / DATA – Output signal (analog for LM35, digital for DHT11)

4-Pin Temperature Sensor (e.g., DHT22)

- VCC – Power supply
- DATA – Digital output
- NC (Not Connected) – No use
- GND – Ground

Digital Sensors (I2C type like DS18B20 sometimes 3 pins)
VCC – Power
GND – Ground
DATA – One-wire communication (requires pull-up resistor)

KEY FEATURES

Accuracy: Indicates how close the measured temperature is to the actual temperature. Higher accuracy sensors give more reliable readings.

Sensitivity: Shows how much the output changes with a small change in temperature. A highly sensitive sensor detects even minor temperature variations.

Range: The minimum and maximum temperatures the sensor can measure (for example, -40°C to 125°C).

Response Time: The time taken by the sensor to detect and reflect a change in temperature. Faster response is important in real-time systems.

Resolution: The smallest change in temperature that the sensor can detect and display.

Stability: Ability to maintain consistent performance over time without drift.

Linearity: Indicates how proportional the output is to the temperature. A linear sensor provides easier and more accurate readings.

Power Consumption: Amount of power required for operation. Low power sensors are preferred for battery-operated and IoT devices.

Output Type: Analog (e.g., LM35),
Digital (e.g., DHT22, DS18B20)

Size and Cost: Compact size and low cost are important for portable and large-scale applications.

3.Pressure Sensor (Potentiometer)



Figure1.4–Pressure Sensor (Potentiometer)

A potentiometer is a three-terminal variable resistor used to control voltage in an electrical or electronic circuit. It works on the principle of a voltage divider, meaning it divides the input voltage into smaller parts based on the position of its adjustable contact called the wiper. The three terminals consist of two fixed ends connected to a resistive track and one movable terminal (wiper) that slides over the track when the shaft is rotated or moved. By changing the position of the wiper, the resistance between the terminals changes, which in turn controls the output voltage. Potentiometers are commonly used to adjust volume in audio devices, control brightness in lights, tune radios, calibrate circuits, and provide input to microcontrollers in embedded systems.

Potentiometers are made of materials such as carbon, cermet, or wire-wound resistive elements. They are available in different types including rotary potentiometers (knob type), linear or slider potentiometers (slide type), and digital potentiometers (electronically controlled). The resistance values typically range from a few ohms to several megaohms depending on the application. When used as a two-terminal device, a potentiometer functions as a rheostat, which controls current rather than voltage. The main advantages of a potentiometer are simple construction, low cost, and easy adjustment, but disadvantages include mechanical wear over time and limited power handling capacity. Overall, a potentiometer is an important and widely used component in electronic circuits for manual control and signal adjustment

PIN OUT OF PRESSURE SENSOR

- VCC – Power supply (usually 3.3V or 5V)
- GND – Ground connection
- OUT (Analog Output) – Gives analog voltage corresponding to pressure

KEY FEATURES

Accuracy: Shows how close the measured pressure is to the actual pressure. Higher accuracy ensures reliable readings.

Pressure Range: The minimum and maximum pressure the sensor can measure (e.g., 0–10 bar, 0–100 psi).

Sensitivity: Indicates how much the output changes for a small change in pressure. Higher sensitivity detects small pressure variations.

Resolution: The smallest change in pressure that the sensor can detect.

Response Time: The time taken by the sensor to respond to pressure changes. Fast response is important in real-time monitoring.

Linearity: Measures how proportional the output signal is to the applied pressure. Better linearity gives more accurate results.

Stability: Ability to maintain consistent readings over time without drift.

Output Type: Analog output (voltage/current),
Digital output (I2C, SPI)

Operating Temperature Range: The range of temperatures in which the sensor works properly.

Power Consumption: Low power consumption is important for IoT and battery-powered

devices.

4.Pin Display OLED



Figure 1.5–Pin Display OLED

A 4-pin OLED display is a small electronic display module commonly used in embedded systems, Arduino, ESP32, and IoT projects to show text, numbers, graphics, and sensor data. The term "4-pin" usually means the display uses I²C communication, which requires only four connections: VCC, GND, SDA, and SCL. OLED stands for Organic Light Emitting Diode, and unlike LCDs, OLED displays do not require a backlight because each pixel produces its own light. This makes the display brighter, thinner, more power-efficient, and capable of showing high contrast, especially for white or blue text on a black background. The four pins have specific functions: VCC is the power supply (usually 3.3V or 5V), GND is ground, SDA (Serial Data) is used to send data to the display, and SCL (Serial Clock) synchronizes the data transmission. Since it uses I²C protocol, only two communication wires (SDA and SCL) are needed, which reduces the number of GPIO pins required from the microcontroller. Most 4-pin OLED displays use the SSD1306 driver IC, which controls the pixels and allows communication with microcontroller

KEY FEATURES

I2C Interface: Uses only 4 pins (VCC, GND, SDA, SCL), reducing wiring complexity and saving microcontroller pins.

Low Power Consumption: Consumes less power compared to LCD displays, suitable for battery-based and IoT applications.

High Contrast Display: Provides very clear and sharp text/images with deep blacks.

Self-Emitting Pixels: No need for a backlight, as each pixel produces its own light.

Wide Viewing Angle: Display can be seen clearly from different angles without color distortion.

Fast Response Time: Quickly updates data, useful for real-time monitoring systems.

Compact and Lightweight: Small size (commonly 0.96 inch) makes it ideal for embedded systems.

Good Resolution: Typically available in 128×64 or 128×32 pixels for clear output.

Easy Interfacing: Works easily with Arduino, ESP32, Raspberry Pi using libraries like SSD1306.

5.Buzzer



Figure1.6–Buzzer

There are two main types of buzzers: active buzzer and passive buzzer. An active buzzer has a built-in oscillator circuit, so it produces sound automatically when a DC voltage (usually 3V to 5V) is applied. It is simple to use because it only requires power supply to generate a beep sound. In contrast, a passive buzzer does not have an internal oscillator and requires an external signal (square wave or PWM signal) from a microcontroller to generate sound. This allows control over different frequencies and tones, making it suitable for music or variable sound alerts.

Buzzers are generally small, lightweight, low-cost, and consume very little power. They typically have two pins: positive (VCC) and negative (GND). The sound is produced by the vibration of a diaphragm inside the buzzer. In piezoelectric buzzers, a piezo crystal vibrates when voltage is applied, creating sound waves. The advantages of buzzers include simple operation, low power consumption, and reliable performance. However, they produce limited sound levels and simple tones compared to speakers. Overall, a buzzer is an important output device in electronic and embedded systems for providing audible alerts and notifications.

KEY FEATURES

Sound Generation: Produces an audible sound (beep or tone) used for alerts, alarms, and notifications.

Types: Active buzzer – Has built-in oscillator, produces sound when power is applied,
Passive buzzer – Requires an external signal to generate sound

Operating Voltage: Typically works at 3.3V or 5V, suitable for microcontrollers like Arduino and ESP32.

Low Power Consumption: Consumes very little power, making it ideal for portable devices.

Compact Size: Small and lightweight, easy to integrate into circuits.

Sound Intensity: Produces sound in a specific range (measured in decibels, dB).

Frequency Rang: Can generate sound at different frequencies (especially passive buzzers).

Easy Interfacing: Can be directly connected to digital pins of microcontrollers.

Quick Response: Instantly produces sound when triggered.

Cost-Effective: Low cost and widely available.

6.Connecting Wires

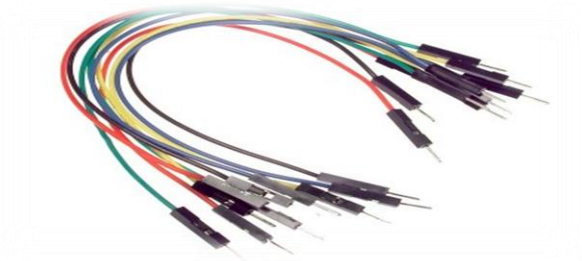


Figure1.7–Connecting Wires

In electronics and prototyping, especially with Arduino, ESP32, and other development boards, connecting wires are commonly called jumper wires. These wires are available in three main types: male-to-male, male-to-female, and female-to-female. Male pins have exposed metal connectors, while female pins have socket-type connectors. These wires are widely used with breadboards to quickly build and test circuits without soldering. In permanent circuits, wires are often soldered onto printed circuit boards (PCBs) for strong and reliable connections

Connecting wires are electrical conductors used to connect different electronic components in a circuit so that current and signals can flow between them. They are very important in electrical, electronic, and embedded systems because without proper wiring, components like sensors, microcontrollers, displays, and power supplies cannot communicate or operate.

CHAPTER-5

CIRCUIT DIAGRAM FIGURE

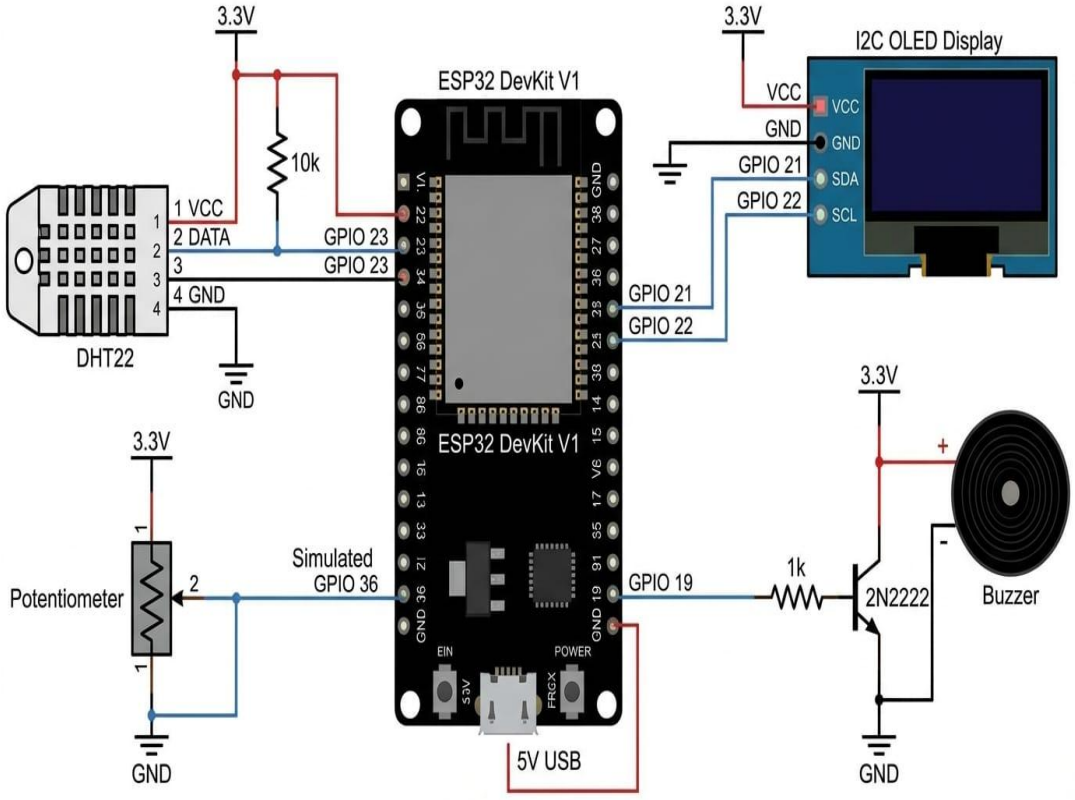


Figure1.8–Circuit diagram

CHAPTER-6

WORKING PRINCIPLE

The working principle of an IoT-based boiler temperature and pressure monitoring system is based on sensing, processing, and transmitting real-time data over the internet for remote monitoring and control. In this system, temperature and pressure sensors are installed on the boiler to continuously measure the operating conditions.

A temperature sensor (such as DHT22, LM35, or thermocouple) measures the heat level inside the boiler, while a pressure sensor measures the steam or internal pressure generated. These sensors convert physical parameters (temperature and pressure) into electrical signals.

The sensed signals are then sent to a microcontroller such as Arduino or ESP32. If the sensors give analog output, the microcontroller converts it into digital form using an ADC (Analog-to-Digital Converter).

The microcontroller processes the data, compares it with predefined safety limits, and prepares it for transmission. If the temperature or pressure exceeds the safe limit, the controller can activate an alert system like a buzzer or send warning notifications.Using a Wi-Fi or IoT communication module, the processed data is transmitted to a cloud platform or IoT server.

The data can be viewed remotely through a mobile app or web dashboard in real time. This allows operators to monitor boiler conditions from anywhere. The system may also store data in the cloud for analysis and maintenance planning. In advanced systems, automatic control actions can be taken, such as switching off the boiler or releasing pressure if unsafe conditions are detected

CHAPTER-7

OPERATION

An IoT-based boiler temperature and pressure monitoring system is designed to continuously measure and monitor the temperature and pressure of a boiler using sensors and send the data to the internet for real-time monitoring. This system helps improve safety, efficiency, and automation in industrial boilers.

The operation of an IoT-based boiler temperature and pressure monitoring system involves continuous sensing, processing, and remote monitoring of critical boiler parameters to ensure safe and efficient performance. In this system, temperature and pressure sensors such as the DHT22 (for temperature) and a pressure sensor are installed on the boiler to measure real-time conditions.

These sensors collect data and send it to a microcontroller like the ESP32, which acts as the central processing unit. The ESP32 reads the analog or digital signals from the sensors, converts them into meaningful values, and processes them based on predefined threshold limits.

The measured temperature and pressure values are then displayed locally on a 4-pin OLED display for immediate observation. Simultaneously, the ESP32 uses its built-in Wi-Fi capability to transmit the data to an IoT cloud platform or mobile application, allowing remote monitoring from anywhere. If the temperature or pressure exceeds safe limits, the system triggers an alert mechanism such as a buzzer to warn operators and prevent potential hazards like overheating or explosion.

This real-time monitoring and alert system improves safety, reduces manual supervision, and enables quick decision-making. Overall, the integration of sensors, microcontroller, display unit, and cloud connectivity ensures efficient and reliable operation of the boiler system.

CHAPTER-8

8.0 IMPLEMENTATION AND TESTING

Implementation:

The implementation of an IoT based boiler temperature and pressure monitoring system involves the integration of sensors, a microcontroller, and an internet communication module to continuously monitor the boiler conditions.

First, the temperature sensor (DHT22) and pressure sensor are connected to the ESP32 microcontroller module. The ESP32 acts as the main controller which collects data from the sensors. The sensors measure the temperature and pressure inside the boiler and send the data to the microcontroller in digital form.

The ESP32 processes the sensor data and sends it to an IoT cloud platform through a Wi-Fi network. The collected data can be viewed in real time on a mobile application or web dashboard. If the temperature or pressure exceeds the safe limit, the system triggers an alert using a buzzer or notification to warn the operator.

Additional components such as OLED display can be used to show the current temperature and pressure values locally. The system is powered through a regulated power supply, and all components are connected using proper interfacing circuits.Thus, the implementation enables real-time monitoring, remote access, and safety control of the boiler system

Testing: Testing is performed to ensure that the system works accurately and reliably.

1.Sensor Testing: The temperature and pressure sensors are tested individually to verify that they measure values correctly.

2.Microcontroller Testing: The ESP12E module is programmed and tested to ensure it reads the sensor data and processes it properly.

3.Communication Testing: The Wi-Fi connection and IoT platform are tested to confirm that data is successfully transmitted to the cloud and displayed on the dashboard.

4.Alert System Testing: The system is checked by setting threshold limits. When the temperature or pressure crosses the limit, the buzzer or notification should activate.

5.System Integration Testing: Finally, the complete system is tested together to ensure all components work smoothly and provide continuous monitoring

1.1 MERITS

1. Real-Time Monitoring : Continuously tracks temperature and pressure levels of the boiler, Operators can view data instantly on mobile apps or web dashboards.

2. Improved Safety : Detects abnormal temperature or pressure early, Sends automatic alerts or alarms to prevent boiler explosions or failures.

3. Remote Access: Engineers can monitor the boiler from any location using the internet, Reduces the need for constant physical inspection.

4. Data Logging and Analysis : Stores historical data for performance analysis, Helps identify trends and predict possible faults.

5. Preventive Maintenance : Early detection of issues allows timely maintenance, Reduces downtime and repair costs.

6. Increased Efficiency : Helps maintain optimal temperature and pressure levels, Improves overall **y**. boiler performance and energy efficiency

1.2 DEMERITS

1.High Initial Cost: The system requires sensors, microcontroller modules, internet connectivity, and cloud platforms. This increases the initial installation cost.

2.Internet Dependency : The system depends on a stable internet connection. If the internet fails, real-time monitoring and data transmission may stop.

3.Security Issues : Since the system is connected to the internet, there is a risk of cyber attacks or unauthorized access to the monitoring data.

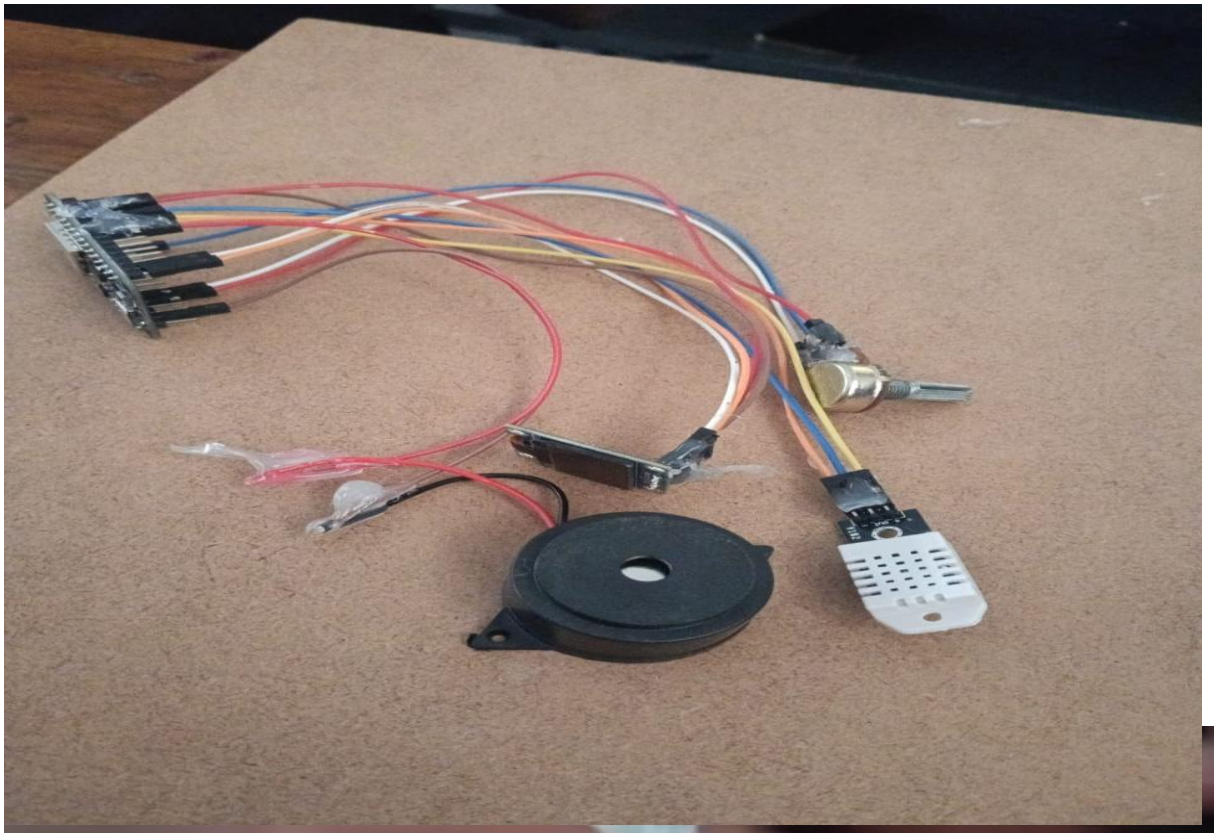
4.Maintenance Requirement : Sensors and electronic components require regular maintenance and calibration to maintain accuracy.

5.Power Supply Dependency : The system needs continuous power. Power failure may interrupt the monitoring process.

6.Complex System Design : Compared to traditional monitoring systems, IoT systems require knowledge of programming, networking, and hardware integration.

CHAPTER-9

9.0 RESULT





1.1 APPLICATIONS

1.Power Plants : Used in thermal power plants to continuously monitor boiler temperature and pressure to ensure safe and efficient power generation.

2.Industrial Manufacturing : Many industries such as chemical, textile, paper, and food processing use boilers. IoT monitoring helps maintain proper operating conditions.

3.Oil and Gas Industry : Boilers used in refineries require constant monitoring to prevent overheating or pressure accidents.

4.Pharmaceutical Industry: Used to maintain accurate temperature and pressure condition equired for sterilization and production processes.

5.Food Processing Industry : Boilers are used for cooking, sterilization, and heating. IoT systems help monitor and control the process remotely.

6.Remote Monitoring Systems : Operators can monitor boiler parameters from anywhere using mobile phones

9.2 PREVENTION TECHNIQUES

1.Setting Safe Threshold Limits : The system should be programmed with maximum safe limits for temperature and pressure. If the values exceed the limit, the system generates an alert or activates a safety mechanism.

2.Regular Sensor Calibration : Temperature and pressure sensors must be calibrated periodically to ensure accurate readings and avoid false alarms.

3.Automatic Alert System: Notifications or alarms (buzzer/mobile alerts) should be used to inform operators immediately when abnormal conditions occur.

4.Overpressure Safety Valves : Installing pressure relief valves in the boiler helps release excess pressure and prevents boiler explosion.

5.Continuous Monitoring through IoT : Real-time monitoring using IoT allows operators to track boiler conditions remotely and take quick action when necessary.

CHAPTER-10

10.0 CONCLUSION

The IoT based boiler temperature and pressure monitoring system is an advanced and efficient solution for monitoring the operating conditions of industrial boilers. Boilers play an important role in many industries such as power plants, chemical industries, food processing units, and manufacturing plants. Maintaining proper temperature and pressure levels inside the boiler is very important for safe and efficient operation. Traditional boiler monitoring methods require manual observation, which may lead to human errors and delayed responses in critical situations.

The proposed IoT based monitoring system overcomes these limitations by using sensors, microcontrollers, and internet connectivity. In this system, temperature and pressure sensors continuously measure the boiler parameters and send the data to the ESP12E microcontroller. The microcontroller processes the data and transmits it to an IoT cloud platform through a Wi-Fi network. The collected data can be accessed in real time using a mobile phone, computer, or web dashboard, allowing operators to monitor the boiler from anywhere.

One of the major advantages of this system is real-time monitoring and quick alert generation. When the temperature or pressure exceeds the predefined safety limits, the system immediately sends notifications or activates alarms such as a buzzer. This helps the operator to take quick action and prevent dangerous situations such as boiler overheating or explosion. The system also improves operational efficiency by providing continuous data monitoring and reducing the need for manual checking.In addition, the IoT based monitoring system helps in data storage and analysis. The collected data can be stored in the cloud for future reference and analysis.

10.1 REFERENCE

1. Raj Kamal, Embedded Systems: Architecture, Programming and Design, Tata McGraw-Hill Education, New Delhi.

2.Vijay Madisetti and Arshdeep Bahga, Internet of Things: A Hands-On Approach, Universities Press, 2014.

3.Adrian McEwen and Hakim Cassimally, Designing the Internet of Things, Wiley Publications.

4.K. V. K. K. Prasad, Embedded/Real-Time Systems: Concepts, Design and Programming, Dreamtech Press.

5.ESP8266 (ESP32) Datasheet, Espressif Systems, Available at: https://www.espressif.com

6.DHT22 Temperature and Humidity Sensor Datasheet, Aosong Electronics.

7.Research papers from IEEE Xplore Digital Library related to IoT based industrial monitoring systems.

8.Articles and tutorials from IoT development websites such as Arduino.cc and RandomNerdTutorials.com.

10.2 APPENDICES

```
#include <ESP8266WiFi.h>
#include <DHT.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// OLED display size
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT,&Wire, OLED_RESET);

// DHT22 settings
#define DHTPIN D4
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);

// Potentiometer & Buzzer
#define POT_PIN A0
#define BUZZER_PIN D5

void setup() {
  Serial.begin(9600);

  dht.begin();
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED not found");
```

```
    while (true);
  }

  display.clearDisplay();
  display.setTextColor(WHITE);
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  int potValue = analogRead(POT_PIN);

  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("DHT read failed");
    return;
  }

  // Buzzer control
  bool buzzerState = false;
  if (potValue > 300) {
    digitalWrite(BUZZER_PIN, HIGH);
    buzzerState = true;
  } else {
    digitalWrite(BUZZER_PIN, LOW);
  }

  // OLED display
  display.clearDisplay();
  display.setTextSize(1);
  display.setCursor(0, 0);

  display.println("DHT + POT + BUZZ");
  display.println("----------------");
```

```
  display.print("Temp: ");
  display.print(temperature);
  display.println(" C");

  display.print("Humidity: ");
  display.print(humidity);
  display.println(" %");

  display.print("Pot: ");
  display.println(potValue);

  display.print("Buzzer: ");
  display.println(buzzerState ? "ON" : "OFF");

  display.display();

  delay(1000);
}
```